

10Rocket-Fuel-cost-benefit

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1 Data Science: Bridging Principles and Practice

1.1 Part 10: Rocket Fuel Costs, Benefits, and Efficacy

In this notebook, we will apply what we learned about Python and DataFrames to compute the costs, benefits, and return on investment for the Rocket Fuel handbag case study.

1.1.1 Table of Contents

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</ol>
```

```
In [1]: # load the necessary software. THIS CELL MUST BE RUN
import pandas as pd
from gofer.ok import check
%matplotlib inline
```

1.2 Rocket Fuel Ad Campaign

1.2.1 (Review from notebooks 03, 04, and 05)

Rocket Fuel Inc. (NASDAQ: FUEL), works in digital advertising offering a "Programmatic Marketing Platform" that claims to optimize digital marketing through big data and machine learning techniques.

In 2015, Rocket Fuel ran a trial ad campaign for handbag manufacturer TaskBella. TaskBella was interested in answering two questions:

1. Would the campaign be successful?
2. If the campaign was successful, how much of that success could be attributed to the ads?

With the second question in mind, they agreed to run an **A/B test**. The majority of the people exposed to Rocket Fuel's content delivery network would see TaskBella's handbag ad (the **experimental group**). But, a small portion of people (the **control group**) would instead see a Public Service Announcement (PSA) in the exact size and place the ad would normally be. One PSA example is below:

1.3 10. Problem: Rocket Fuel Costs, Benefits, and Efficacy

Earlier in the course we explored the Rocket Fuel study in terms of grouping and visualizing the data. Now, we want to see whether or not the campaign worked.

In this section, we'll explore four questions:

- *Was the campaign effective? Did more users convert as a result of seeing an ad?*
- *How much more money did TaskBella make as a result of running the campaign (ignoring advertising costs)?*
- *Was the campaign profitable (what was the ROI)?*
- *What was the opportunity cost of including a control group? How much more could have TaskBella made with a smaller control group or not having a control group at all?*

Run the next cell to load the data. Then, run the cell after it to split the data into two smaller DataFrames: one that only has subjects in the control ("psa") group, and another that only has subject in the test ("ad") group.

```
In [2]: # run this cell to load the data
ads = pd.read_csv('data/rocketfuel_data_renamed.csv', index_col=0)

# show the first five rows of the ads data
ads.head()
```

```
Out[2]:
```

	user id	test group	converted	total ads	most ads	ads day	most ads	hour
0	1069124	ad	False	130		Monday		20
1	1119715	ad	False	93		Tuesday		22
2	1144181	ad	False	21		Tuesday		18
3	1435133	ad	False	355		Tuesday		10
4	1015700	ad	False	276		Friday		14

```
In [3]: # create a DataFrame with only users in the control (PSA) group
control = ads[ads["test group"] == "psa"]

# create a DataFrame with only users in the experimental (ad) group
experiment = ads[ads["test group"] == "ad"]
experiment.head()
```

```
Out[3]:
```

	user id	test group	converted	total ads	most ads	ads day	most ads	hour
0	1069124	ad	False	130		Monday		20
1	1119715	ad	False	93		Tuesday		22
2	1144181	ad	False	21		Tuesday		18
3	1435133	ad	False	355		Tuesday		10
4	1015700	ad	False	276		Friday		14

1.3.1 10a. Did more users convert as a result of the ad campaign?

We're interested in seeing if the buying behavior of users differed between the control and experimental groups. The two groups are very different in size, so it isn't fair to compare the number of people who converted in each group. Instead, we're going to look at the *proportion* of people in each group who bought a bag.

For both groups, the proportion will be calculated as:

$$\frac{\text{number of people in group who converted}}{\text{total number of people in group}}$$

Let's start with the control group. To get the number of people in the control group, we can use the shape attribute to get the shape of the DataFrame.

```
In [4]: control.shape
```

```
Out[4]: (23524, 6)
```

The shape attribute returns two values: the number of rows followed by the number of columns. We only want the number of rows (the first value), so we'll use indexing to get it. Remember that in Python indexing starts at 0.

```
In [5]: # number of users in control group
num_control = control.shape[0]
num_control
```

```
Out[5]: 23524
```

Next, we need a table with only users in the control group who converted. We can get this using Boolean indexing. Our condition inside the square brackets (i.e. how we want to select the rows) is that we want rows where the "converted" feature is True.

```
In [6]: # table with only converting control group users
ctrl_converts = control[control["converted"] == True]
ctrl_converts.head()
```

```
Out[6]:
```

	user id	test group	converted	total ads	most ads day	most ads hour
957	915026	psa	True	88	Sunday	19
2531	904434	psa	True	7	Tuesday	20
2744	920737	psa	True	77	Monday	13
2751	904131	psa	True	111	Monday	12
2979	901749	psa	True	364	Friday	21

From this new table, we can get the number of converting control group users by again using shape.

```
In [7]: # number of people in the ctrl_converts table
num_ctrl_converts = ctrl_converts.shape[0]
num_ctrl_converts
```

```
Out[7]: 420
```

Finally, we can plug the number of control group converts and the total number of control group people into our formula to find the percentage.

```
In [8]: # proportion of control group users who converted
        ctrl_convert_proportion = (num_ctrl_converts / num_control)
        ctrl_convert_proportion
```

```
Out[8]: 0.01785410644448223
```

EXERCISE: Find the proportion of people in the experiment group who converted. You can follow the exact same steps as we did above for the control group; in all steps the code will be identical except for the variable and table names.

Step 1: Get the number of people in the experiment group using the experiment table, the shape attribute, and indexing.

```
In [9]: # number of people in the experiment (ad) group
        num_exper = experiment.shape[0]
        num_exper
```

```
Out[9]: 564577
```

Step 2: Fill in the ellipses with the correct condition to select users in the experiment group who converted.

Hint: if you're stuck, look at how we did it for the control group. It's the same task, so the code will look very similar, but all references to the control DataFrame will be replaced by the experiment DataFrame.

```
In [10]: # get only the experiment group users who converted
         exper_converts = experiment[experiment["converted"] == True]
         exper_converts.head()
```

```
Out[10]:
```

	user id	test group	converted	total ads	most ads day	most ads hour
15	1461774	ad	True	9	Wednesday	18
44	1355531	ad	True	265	Tuesday	12
107	1389878	ad	True	1328	Saturday	19
121	1475989	ad	True	323	Saturday	20
135	1241733	ad	True	246	Friday	20

Step 3: Get the number of converted experiment group users using the table you just created, the shape attribute, and indexing.

```
In [11]: # count the number of converting experimental group members
         num_exper_converts = exper_converts.shape[0]
         num_exper_converts
```

```
Out[11]: 14423
```

Step 4: Plug the values from step 1 and step 3 into the formula to calculate the proportion.

$$\frac{\text{number of people in group who converted}}{\text{total number of people in group}}$$

Hint: you don't have to type any numbers here; use the names of the two variables you just created.

```
In [12]: # the proportion of people in the experimental group that converted
exper_convert_proportion = (num_exper_converts / num_exper)
exper_convert_proportion
```

```
Out[12]: 0.025546559636683747
```

```
In [13]: # run this cell to check your answer for common errors
check("tests/exper-convert.ok")
```

```
Out[13]: <gofer.ok.OKTestsResult at 0x7fe215ff4438>
```

The next cell will print the values you calculated as percents of the control and experiment groups that converted, rounded to two decimal places.

```
In [14]: print("Control Group: {} % converted".format(round(ctrl_convert_proportion * 100, 2)))
print("Experiment Group: {} % converted".format(round(exper_convert_proportion * 100,
```

```
Control Group: 1.79 % converted
```

```
Experiment Group: 2.55 % converted
```

QUESTION: Was the campaign effective? Was a user who saw the ad more likely to buy a bag than a user who didn't see the ad?

ANSWER: *Fill in your answer here* Certainly the % conversion for the experiment group is high. We need to check if the difference is statistically significant

1.3.2 10b. How much more money did TaskBella make as a result of running the campaign (ignoring advertising costs)?

Here we're looking for the benefit of the campaign: the expected financial impact from the conversions resulting from the ads (excluding all advertising costs).

The formula for the benefit is as follows:

$(\text{value of a converted user}) * (\text{number of users in the experiment group}) * (\text{proportion of converting experiment group users} - \text{proportion of converting control group users})$

That is, we are looking for the number of people in the experiment group who bought a handbag and *wouldn't have bought one if they'd been in the control group*- the people whose conversion was the result of the ad campaign, This is why we subtract the control group conversion percentage from the experiment group conversion percentage.

We already have most of the parts of this formula- we just need to assemble them.

EXERCISE: TaskBella estimates the value of a converted user to be \$40. In the following cell, assign 40 to the name `convert_val`.

```
In [15]: # dollar value of converted user
convert_val = 40
```

Next, let's get the difference in conversion proportions for the experiment and control groups: proportion of converting experiment group users – proportion of converting control group users. You can do this easily by using the variables you just calculated: `exper_convert_proportion` and `ctrl_convert_proportion`.

```
In [16]: # the difference between the experiment conversion proportion and the control conversion
proportion_diff = exper_convert_proportion - ctrl_convert_proportion
proportion_diff
```

```
Out[16]: 0.007692453192201517
```

Lastly, plug all the appropriate values into the benefit formula to get the benefit.

(value of a converted user) * (number of users in the experiment group) * (proportion of converting experiment

Hint: the number of users in the experiment group is saved as num_exper.

```
In [17]: benefit = convert_val * num_exper * proportion_diff
benefit
```

```
Out[17]: 173719.28583574222
```

```
In [18]: # run this cell to check your answer for some common errors
check("tests/benefit.ok")
```

```
Out[18]: <gofer.ok.OKTestsResult at 0x7fe2147b0400>
```

1.3.3 10c. What was the Return on Investment (ROI)?

In parts 10a and 10b we saw that advertising resulted in a higher percentage of converting users and a positive benefit. But, would using the campaign still increase profits when advertising costs are accounted for?

Recall that back in Notebook 02 we calculated the advertising costs and named them cost. Run the next cell to re-define that variable with the cost we calculated.

```
In [19]: # the cost of the campaign
cost = 131374.64
```

EXERCISE: Calculate the ROI as

$$\frac{\text{benefit} - \text{cost}}{\text{cost}}$$

```
In [20]: # calculate the ROI
# remember to mind your order of operations
roi = (benefit - cost) / cost
roi
```

```
Out[20]: 0.32231978588669935
```

```
In [21]: # run this cell to check your answer for some common errors
check("tests/roi.ok")
```

```
Out[21]: <gofer.ok.OKTestsResult at 0x7fe1e80f7668>
```

1.3.4 10d. What was the opportunity cost of including a control group?

As we saw in 10b, having a control group is important to get a baseline with which to compare the experimental data. However, any users assigned to the control group are not seeing TaskBella's advertising, eating into profits.

We can calculate the *opportunity cost* of the control group as:

$(\text{value of converted user}) * (\text{number of users in control group}) * (\text{proportion of experiment group users who converted})$

In other words, the opportunity cost is the additional amount of money users in the control group would have spent if they had seen the ads *purely as a result of seeing the ads*. Note that this is almost the same formula as for the benefit in 10b, except with the control group instead of the experiment group.

EXERCISE: Use `convert_val`, `num_control`, and `proportion_diff` to calculate the opportunity cost.

```
In [22]: # calculate the opportunity cost
        opp_cost = convert_val * num_control * proportion_diff
        opp_cost
```

```
Out[22]: 7238.2907557339395
```

```
In [23]: # run this cell to check your answer for some common errors
        check("tests/opp-cost.ok")
```

```
Out[23]: <gofer.ok.OKTestsResult at 0x7fe1e8178e80>
```

QUESTION: Was the ad campaign profitable when all the costs are accounted for? Why or why not?

ANSWER: *Fill in your answer here* Yes - positive ROI

References

- Rocket Fuel data and discussion questions adapted from materials by Zsolt Katona and Brian Bell, BerkeleyHaas Case Series

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